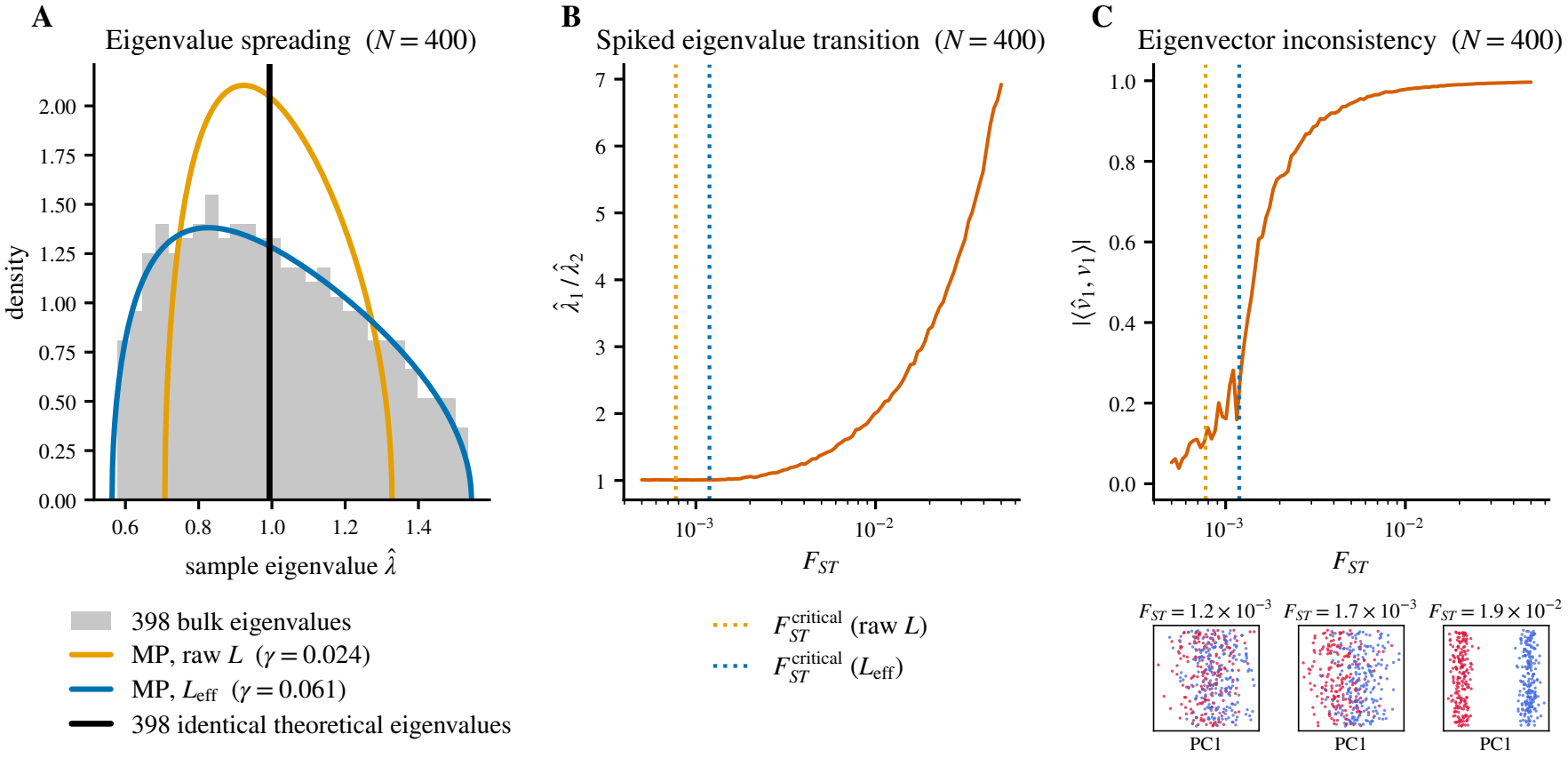
A
Eigenvalue spreading ($N = 400$)
density
sample eigenvalue $\hat{\lambda}$
398 bulk eigenvalues
MP, raw L ($\gamma = 0.024$)
MP, L_{eff} ($\gamma = 0.061$)
398 identical theoretical eigenvalues
B
Spiked eigenvalue transition ($N = 400$)
$\hat{\lambda}_1 / \hat{\lambda}_2$
F_{ST}
F_{ST}^{critical} (raw L)
F_{ST}^{critical} (L_{eff})
C
Eigenvector inconsistency ($N = 400$)
$|\langle \hat{v}_1, v_1 \rangle|$
F_{ST}
$F_{ST} = 1.2 \times 10^{-3}$
$F_{ST} = 1.7 \times 10^{-3}$
$F_{ST} = 1.9 \times 10^{-2}$
PC1
PC1
PC1